

NOEL VINCE C. NUEVE

Software Developer

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OBJECTIVE

To join a growing organization as a Software Developer, where I can apply my technical foundation in software development, embedded systems, and computer vision, while continuing to build my skills and contribute meaningfully to the team's projects and growth.

PROFESSIONAL SUMMARY

Computer Engineering graduate with hands-on experience developing software for real-time embedded and computer-vision systems using Python and C++. Experienced in system integration, multithreaded programming, debugging, computer vision, and hardware-software communication. Built and integrated software across Arduino, ESP32, and Raspberry Pi platforms, with a growing focus on professional software development and building reliable applications.

EXPERIENCE

Technical Support Intern

Apr - Jun 2026

Universal Access & Systems Solutions (UAS), Pasig City

- Diagnosed and resolved network and systems issues using structured troubleshooting methods applicable to software debugging and technical problem-solving.
- Applied systematic root-cause analysis in hands-on network configuration labs (Cisco Packet Tracer), building a disciplined approach to isolating and fixing technical issues — a skill directly transferable to debugging software systems.
- Prepared and delivered a technical presentation to supervisors, demonstrating clear communication of technical concepts to non-technical stakeholders.

FEATURED PROJECTS

Tela-Sorb: Semi-Automated Machine for Sound-Absorbing Panel Production Using Textile Waste

Undergraduate Thesis / Capstone, 4-member team

- Achieved 90-100% color-sorting accuracy across 150 test trials, with the detection model reaching 0.995 mAP@0.5 and a 0.98 F1 score, exceeding the project's own 90%+ accuracy target.
- Designed and built a semi-automated system that converts textile waste into acoustic panels, integrating load-cell weighing, computer vision-based color sorting, and compression molding to produce a uniform 7×7-inch panel from 160g of textile scraps.
- Addressed textile waste accumulation while offering an affordable, eco-friendly alternative to commercial acoustic materials for homeowners, students, and small studios.

My Contribution:

- Programmed the core logic connecting the Arduino Nano, ESP32, and Raspberry Pi 5, enabling real-time coordination between the computer vision, weighing, and actuator subsystems.
- Wired key hardware components and developed the Python/C++ software for YOLO-based color detection and multi-threaded sensor and actuator coordination.

Chexam: An Offline Automated Exam Checker Device for Multiple Choice, True or False, Identification, and Essay

Elective Course Project, 4-member team

- Trained custom YOLO (You Only Look Once) models to recognize handwritten letters, achieving 96-100% detection accuracy across multiple choice, true/false, identification, and essay categories.
- Developed an offline device that automatically grades handwritten exams using OCR and YOLO-based detection, with ArUco marker-based alignment for reliable paper detection under varying camera angles and lighting.
- Designed for classrooms with limited or no internet access, reducing manual grading workload for teachers and enabling faster, more consistent feedback for students.

My Contribution:

- Programmed the core software: image capture, OCR/YOLO-based handwriting recognition, answer-key comparison, and scoring logic.
- Trained and fine-tuned the YOLO model on custom-labeled datasets, and integrated the software with the system's hardware modules (camera, IR sensor, feeder motor, thermal printer).

Additional projects available upon request.

TECHNICAL SKILLS

Programming (Developing): Python, C++, building system logic and hardware communication with growing independent coding ability; also exposed to Java and MATLAB

Currently Learning: JavaScript, HTML, CSS, React, Node.js, working toward full-stack web development

Computer Vision (Intermediate): OpenCV, YOLO, object labeling, model training, and accuracy tuning for real-time detection

Embedded & Hardware (Proficient): Arduino, ESP32, Raspberry Pi, I2C, serial communication, sensor-to-microcontroller wiring

Other Tools: Arduino IDE, Git (basic), TCP/IP & REST concepts, Cisco Packet Tracer

Core Competencies: Debugging, Problem-Solving, System Integration, Technical Documentation, Cross-Functional Collaboration

EDUCATION

BS Computer Engineering

2022 - 2026

National University, Fairview, Quezon City

TRAINING & CERTIFICATIONS

- CCNA & Cybersecurity Trainee: Experts Academy, Metro Manila | March to May 2022-2026. Completed hands-on configuration and scripting labs covering TCP/IP, subnetting, VLANs, and ACLs.
- Certification: Fortinet NSE1 (Fortinet, 2026).
- Certificate of Participation, AI for Public Good: AI for Impact Prompt Engineering Workshop, Cognizant, August 29, 2025.

AWARDS

- Academic Excellence Award, Top 8 Student: Fourth Year Level, Second Term A.Y. 2025-2026 (GWA: 2.75).
- Dean's Lister, Second Honor: Second Honor, First Term A.Y. 2022-2023 (Rank 1, BS Computer Engineering).